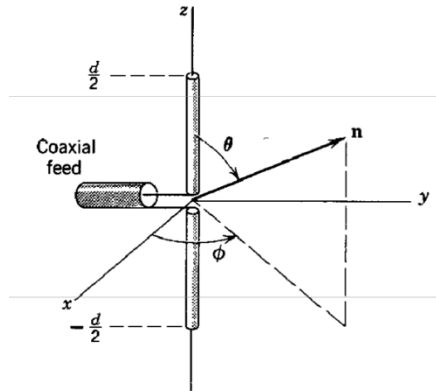


Homework2 Due by 6:00 PM 06/06 Web uploading.

1. (20) Radiation from a center-fed linear antenna



Consider a coaxial center-fed antenna with the current density,

$$\vec{J}(\vec{x}) = I \sin\left(\frac{kd}{2} - k|z|\right) \delta(x)\delta(y)\hat{z}$$

$$|z| < d/2$$

(a) (3) Find the vector potential in the radiation zone

(b) (3) Find the

(c)

$$\frac{dP}{d\Omega} = \frac{Z_0 I^2}{8\pi^2} \left| \cos\left(\frac{kd}{2} \cos\theta\right) - \cos\left(\frac{kd}{2}\right) \right|^2 / |\sin\theta|^2$$

(d) $\frac{dP}{d\Omega} = \frac{Z_0 I^2}{8128} (kd)^2 |\sin\theta|^2$

(e) $kd = \pi$ and $kd = 2\pi$,

(f) (4) Plot the angular distributions for (e).

2. (20) Angular distribution of the multipole radiation

$$\frac{dP(l, m)}{d\Omega} = \frac{Z_0}{2k^2} |a(l, m)|^2 |X_{lm}|^2,$$

with the X_{lm} defined during the class (in the textbook).

(a) (10) Find the explicit form of $|X_{lm}(\theta, \phi)|^2$ for $l = 1$ (dipole) & 2 (quadrupole)

with allowed m 's for each case.

(b) (10) Draw the radiation patterns for pure (l, m) multipoles

J.D. Jackson, Chapter 9 (3rd edition)

3. (20) Prob. 9.1
4. (20) Radiating quadrupole (Prob. 9.2)
5. (20) Radiating two fixed dipoles (Prob. 9.15)